

Rapid Prototyping of Advanced Packaging using 3D-Printed Fanout Interposer Substrates with Diagonal Through-Holes

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Novelty of work

The novel contribution of this paper arises from an innovative approach to fabricating a fanout (FO) interposer substrate by 3D-printing technology. The approach eases the burden of complex fabrication processes of layer-by-layer stacking, etching, and metallization for multiple build-up layers. 3D-printed diagonal through-holes enable interconnection by one-step metallization.

Abstract Text

Artificial intelligence (AI) has carried out notable advances in various fields such as deep learning, automotive, 5G mobile communication, and big data analytics. These advances are driven by advanced packaging which merges multiple components as a single electronic device by an electrical interface, interposer. The interposer is mainly required to obtain higher I/O density for chip-to-chip or chip-on-substrate interconnections [1]. The interconnection spreads the shorter chip-side pad pitch to the larger substrate-side pad pitch by a fanout (FO) structure [2]. However, the fabrication for the FO interconnections comprises build-up layer stacking, photolithography, etching, metallization for via and redistribution layer, and chemical mechanical polishing. Multiple times of the procedure is repeated depending on the number of build-up layers [3]. Recently, packaging work using 3DP technology has been suggested due to its cost-effectiveness and simplified processes [4]. The printing method enables a tilted shape to effectively spread the metal lines.

In this paper, we demonstrated the fabrication of a 3D-printed organic FO interposer substrate, which links the chips through metalized holes without destructive de-smear and etching processes leading to minimized failure during the processes. The experimental procedure lies in the design of an FO substrate with a 50- μm -diameter embedded diagonal through-holes spreading to a larger pitch using a high-resolution projection micro stereolithography (P μ SL) 3D printing technique. A two-micron-thick Cu seed layer was first plated in the holes using electroless plating followed by a hydrophilization process to guide the aqueous plating solution into them. The 2-micron z-axis resolution is a key factor in realizing the metallization of through-holes along with the hydrophilization process, which minimizes the flow resistance originating from the bumpiness of the step-wise walls in angled through-holes. The curved shape was employed to keep the distance between lines evenly and prevent the sharp bend. The metal was filled by electroplating on the metal seed layer with the reverse pulse for uniform growth. The quality of the deposited metal was cross-validated by microscopic images with mechanical milling to observe a cross-sectional view. The interconnection of the metalized 3D-printed organic interposer substrate was characterized by bonding silicon coupon chips. The bonding process was conducted with low-melting point metals (e.g., Bi and In) to reduce the heat damage to the substrate. The alternating bonding junctions and metal lines of the daisy

chain can bring their resistance with varying distances and the number of junctions. Those characterization methods can be applied to the conditions where the chips are bonded on the substrate for chip-to-chip or chip-to-substrate connections.

The introduction of 3D printing technology to interconnect the chips with a one-step distributing FO structure opens the rapid prototyping of advanced packaging technologies. Facile integration and practical operation of destructive chip connections by advanced packaging can be verified with 3D-printed FO interposers at low cost can be demonstrated.

References

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Figures

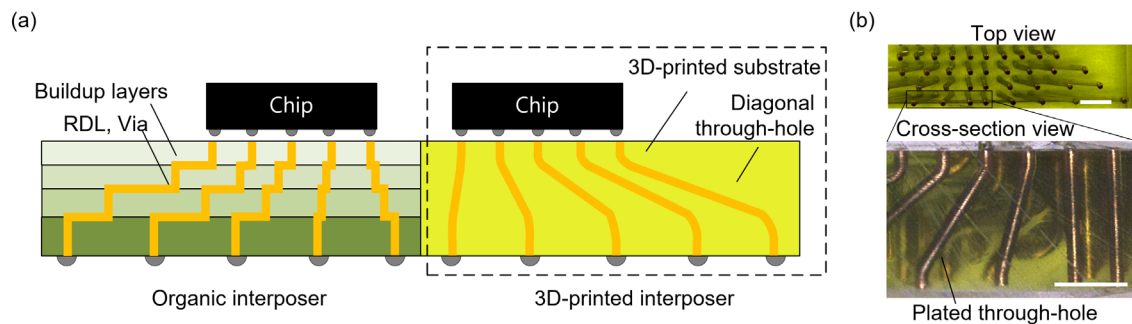


Figure 1. Organic interposer and 3DP interposer. (a) Illustration of interposer between chip and substrate. (b) Image of 3DP interposer with metal lines (scale bar = 500 μm).